Patch GP: Gaussian Process–Based Lossy Compression of Scientific Datasets

Hurricane ISABEL W Field Case Study

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Outline

1. Motivation
2. Background: SZ3 ALGO_INTERP
3. Patch GP Algorithm
4. Experiment Setup
5. Results
6. Future Work

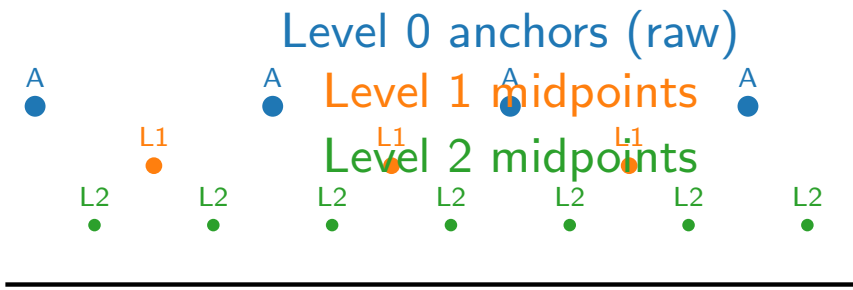
Motivation

- Scientific simulations produce **large floating-point arrays** (e.g. Hurricane ISABEL: $100 \times 500 \times 500 \approx 25$M values).
- Lossy compression under an **absolute error bound** (EB) is the standard regime: every reconstructed value satisfies $|\hat{f}(\mathbf{x}) - f(\mathbf{x})| \leq \mathrm{EB}$.
- State-of-the-art: **SZ3 ALGO_INTERP** — hierarchical dyadic cubic-spline prediction followed by quantisation and entropy coding. Fast and effective, but uses a *fixed predictor*.
- **Our goal:** replace the fixed interpolant with a *Gaussian Process* (GP) that
  1. adapts its predictor to the data's spatial covariance;
  2. selects anchor points (raw-stored values) *optimally* via randomised pivoted Cholesky (RPCholesky);
  3. propagates quantisation uncertainty through the noise model.

SZ3 ALGO_INTERP — Baseline

1. Place coarse anchor grid at stride s (power of 2). Store all anchors as **raw float32**.
2. At each level k: predict midpoints along $X \to Y \to Z$ by cubic-spline interpolation from 4 nearest known values.
3. Quantise residual: $q = \lfloor e \,/\, (2 \cdot \mathrm{EB}) \rceil$, store q (entropy coded).
4. Use *reconstructed* value $\hat{f} + q \cdot 2 \cdot \mathrm{EB}$ as input for subsequent predictions.
5. Repeat until all N values resolved.

Key property: total work $O(N)$; predictor has no free parameters; *no* anchor selection.



Patch GP — Overview

Core idea

Tile the volume into a two-level spatial partition. The *center* of each inner block is an **anchor candidate**. A GP with RPCholesky sensor selection then processes blocks in K rounds, one inner-block position per round.

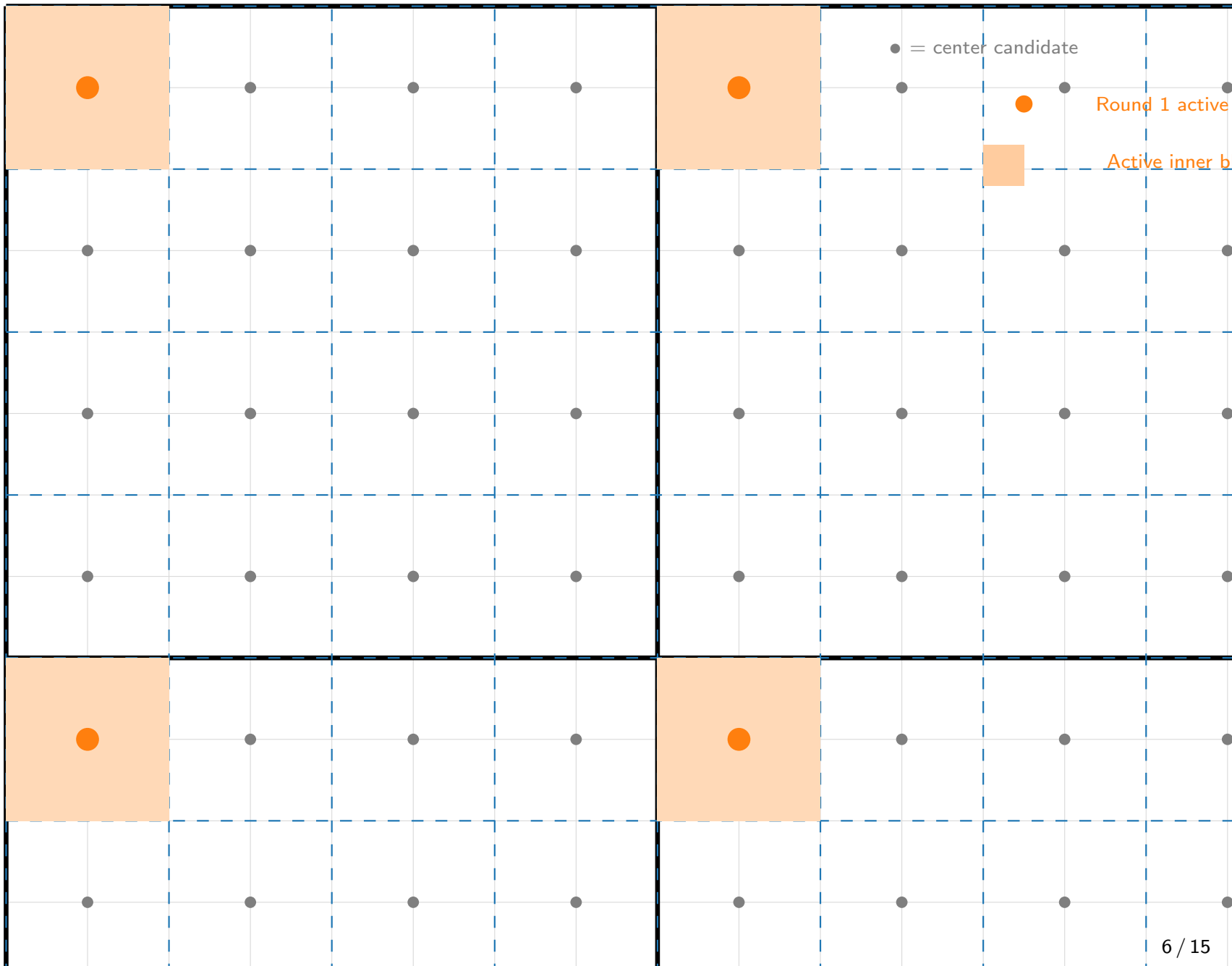
What stays the same as SZ3:

- Absolute error bound EB
- Quantisation: $q = \lfloor e/(2 \cdot \mathrm{EB}) \rceil$
- Reconstructed values feed subsequent predictions
- Entropy coding of all q-bins

What changes:

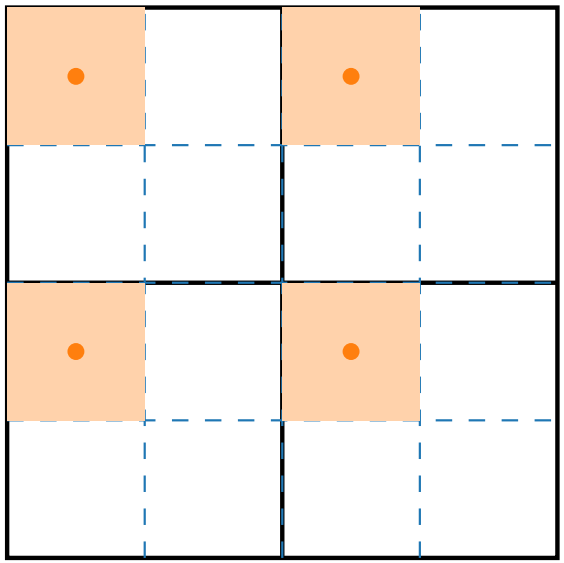
- Predictor: GP (Matérn-5/2, tuned HPs)
- Anchor selection: RPCholesky+GKS (space-filling)
- Noise model: $\sigma_n = \mathrm{EB}/\sigma_y$ for pseudo-anchors
- Processing order: patch-by-patch, not dyadic level

Patch Structure (2-D Illustration)

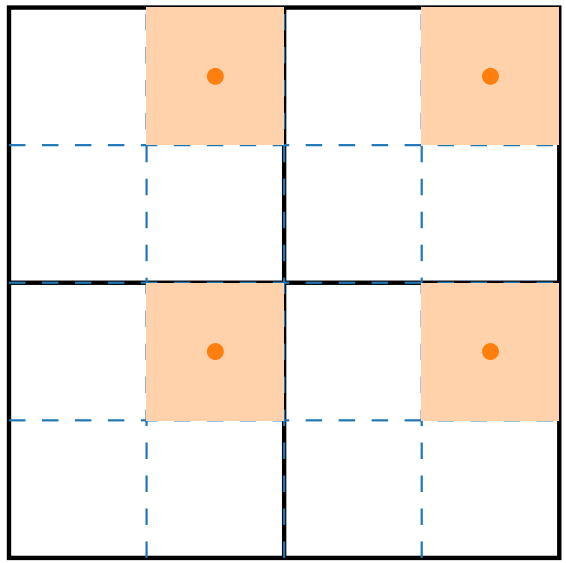


outer

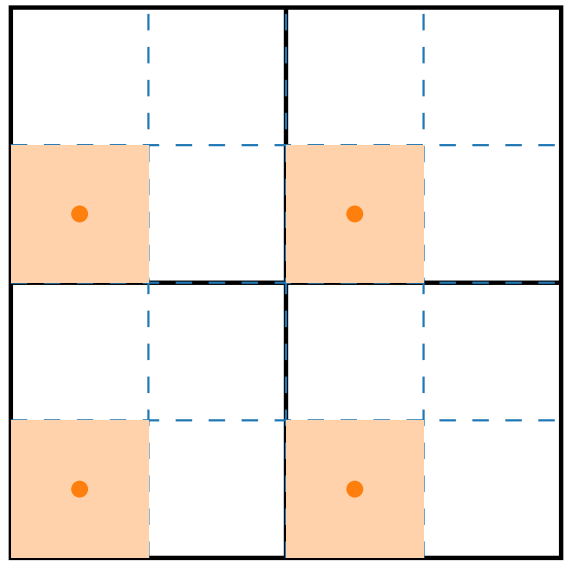
Round-by-Round Processing



Round 1



Round 2



Round 3

- In each round: **same inner-block position** from *every* outer block.
- RPCholesky sees M center candidates *spread across the volume* $\Rightarrow$ good global space-filling selection.
- After K rounds: **all N values resolved**.

Patch GP Algorithm

Notation

M outer blocks; K rounds; $\mathcal{C}_r$ = center candidates in round r; $\mathcal{B}_{r,m}$ = non-center pixels of outer block m in round r.

1. Fit HP $(\ell_{xy}, \ell_z, \sigma^2)$ on $\sim$600 random samples (L-BFGS-B MLE)
2. Compute patch sizes from ℓ_{xy}, ℓ_z (largest divisors $\leq$ one length scale)
3. **For** $r = 1, \ldots, K$:
   - **A. Centers:** RPCholesky selects $\lfloor \alpha M \rceil$ anchors from $\mathcal{C}_r$; store raw float32. GP predicts remaining centers; quantise $q_i = \lfloor e_i/(2\,\mathrm{EB}) \rceil$. Add reconstructed values to conditioning set.
   - **B. Interiors:** GP predicts all $\bigcup_m \mathcal{B}_{r,m}$ (K_LOCAL nearest nbrs). Quantise; add reconstructed values to conditioning set.
4. Compute entropy H of all q-bins; bit rate $= (32\,N_{\mathrm{anc}} + H\,N_{\mathrm{pred}})/N$

Heteroscedastic Noise Model

GP posterior mean

$$\mu(\mathbf{x}^*) = \mathbf{k}^{*\top}\left(\mathbf{K} + \boldsymbol{\Sigma}_n\right)^{-1}\mathbf{y}$$

where $\boldsymbol{\Sigma}_n = \mathrm{diag}(\sigma_{n,i}^2)$ is *per-point* noise.

Point type	$\sigma_{n,i}$
True anchor (float32)	≈ 0
Pseudo-anchor (quantised)	$\mathrm{EB}/\hat{\sigma}_y$

Why this matters:

- Quantised values are known only to $\pm\mathrm{EB}$.
- Propagating that uncertainty into the posterior gives honest confidence intervals.
- SZ3 cannot do this — it has no noise model.
- Larger EB $\Rightarrow$ larger $\sigma_n \Rightarrow$ GP naturally deweights noisy pseudo-anchors.

Patch GP vs SZ3 ALGO_INTERP: Key Differences

	SZ3 ALGO_INTERP	**Patch GP**
Predictor	Cubic spline (4 pts)	GP, Matérn-5/2 (K_LOCAL pts)
Anchor placement	Fixed coarse grid	RPCholesky (space-filling)
Anchor fraction	100% of coarse grid	$\alpha = 10\%$ of candidates
Noise model	None	Heteroscedastic ($\sigma_n = \mathrm{EB}/\sigma_y$)
HP tuning	None	Global MLE ($\sim$600 pts)
Processing order	Dyadic levels (X$\rightarrow$Y$\rightarrow$Z)	Patch rounds
Computational cost	$O(N)$	$O(N \cdot K_{\mathrm{local}}^3)$

Experiment Setup

Dataset

- Hurricane ISABEL simulation (NCAR)
- Field: vertical wind velocity W
- Dimensions: $100 \times 500 \times 500$ ($N = 25{,}000{,}000$ values)
- Format: float32 (≈ 95 MB raw)

Error bounds tested

$$\mathrm{EB} \in \{0.005,\ 0.01,\ 0.02,\ 0.05,\ 0.1\}$$

Patch GP settings

- Kernel: anisotropic Matérn-5/2
- HP fitting: 600 pts, L-BFGS-B
- Anchor fraction $\alpha = 0.10$
- $K_{\mathrm{local}} = 20$ nearest neighbours
- Patch sizes: data-adaptive (largest divisor of 500/100 $\leq \ell \cdot N_{\mathrm{dim}}$)
- Outer mult $= 4$

Baseline

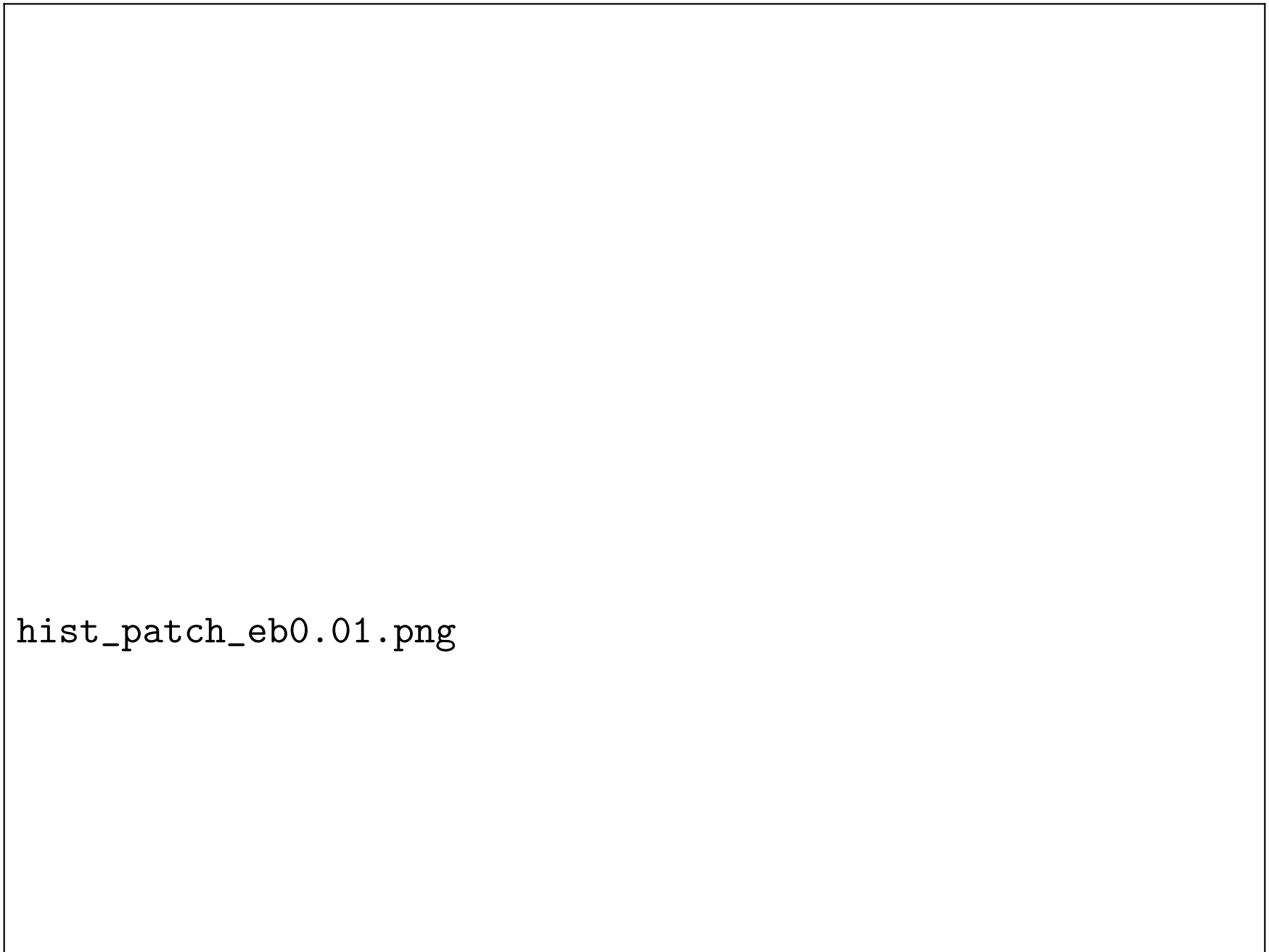
- SZ3 v3 ALGO_INTERP (cubic spline)

Metrics: PSNR (dB), bit rate (bpv), compression ratio (CR).

Rate–Distortion Curves



Quantisation Error Histograms (EB = 0.01)



Future Work

1. **Local HP fitting per round** — use M center candidates as tuning data for round-specific $(\ell_{xy}, \ell_z, \sigma^2)$.
2. **Adaptive patch sizing** — octree decomposition; finer patches in high-variance regions (e.g. hurricane eye).
3. **Boundary continuity** — Patchwork Kriging (Park & Apley, JMLR 2018) to remove discontinuities at patch edges.
4. **Comparison with Progressive GP** — dyadic midpoint version (separate script) on same RD axes.
5. **SZ3 ALGO_LORENZO_REG** as a second baseline.
6. **Other fields** — U, V, pressure, temperature from ISABEL.
7. **Computational scaling** — batch Cholesky across patches; GPU acceleration.

Conclusion

- Patch GP replaces SZ3's fixed cubic-spline predictor with a **data-adaptive Gaussian Process**.
- **RPCholesky** selects space-filling anchors — reducing raw storage relative to SZ3's fixed coarse grid.
- **Heteroscedastic noise** propagates quantisation uncertainty honestly through the GP posterior.
- The two-level spatial partition enables **globally-aware sensor selection** (M candidates spread across the volume) while keeping individual GP solves small.
- RD comparison against SZ3 ALGO_INTERP is in progress.

Key references

Park & Apley (2018). *Patchwork Kriging for Large-scale GP Regression.* JMLR.
Datta et al. (2016). *Hierarchical Nearest-Neighbour GP Models.* JASA.
SZ3: Liang et al. (2022). *SZ3: A modular framework for composing prediction-based error-bounded lossy compressors.* IEEE TPDS.